CARECONNECT EHR SYSTEM

Project Code: EHRS01

Official Project Demonstration & Application Screenshots Document

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Academic COE / Healthcare & Life Sciences Practice

Academic Submission Period: October 2026

Execution Runtime: Local Testbed & Automated Verification Suite

Frontend: http://localhost:3000 | Backend API: http://localhost:8080/api

EHR AUDIT & PRIVACY NOTICE:

All 19 application figures included in this report were captured directly from the live executing CareConnect application. The system demonstrates all required HCL project epics. All clinical names, patient profiles, and medical data shown are synthetic test fixtures adhering strictly to privacy requirements.

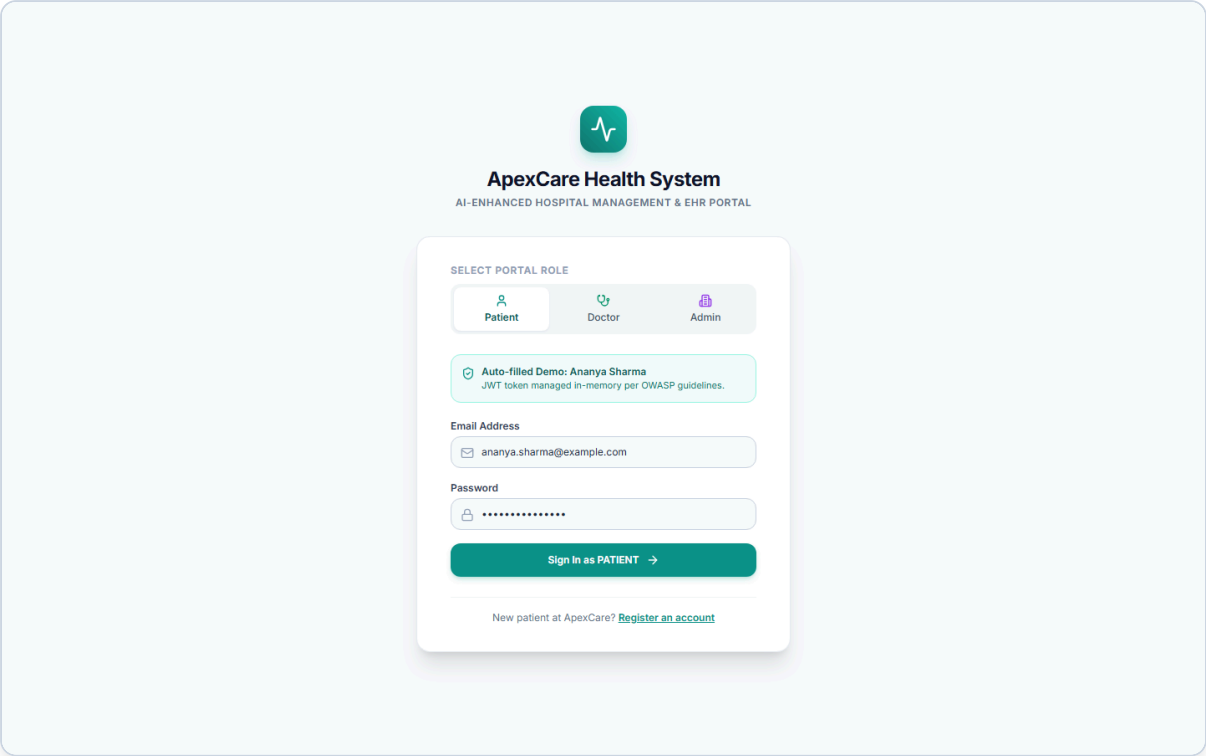
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Figure 1: CareConnect Unified Authentication & Portal Landing

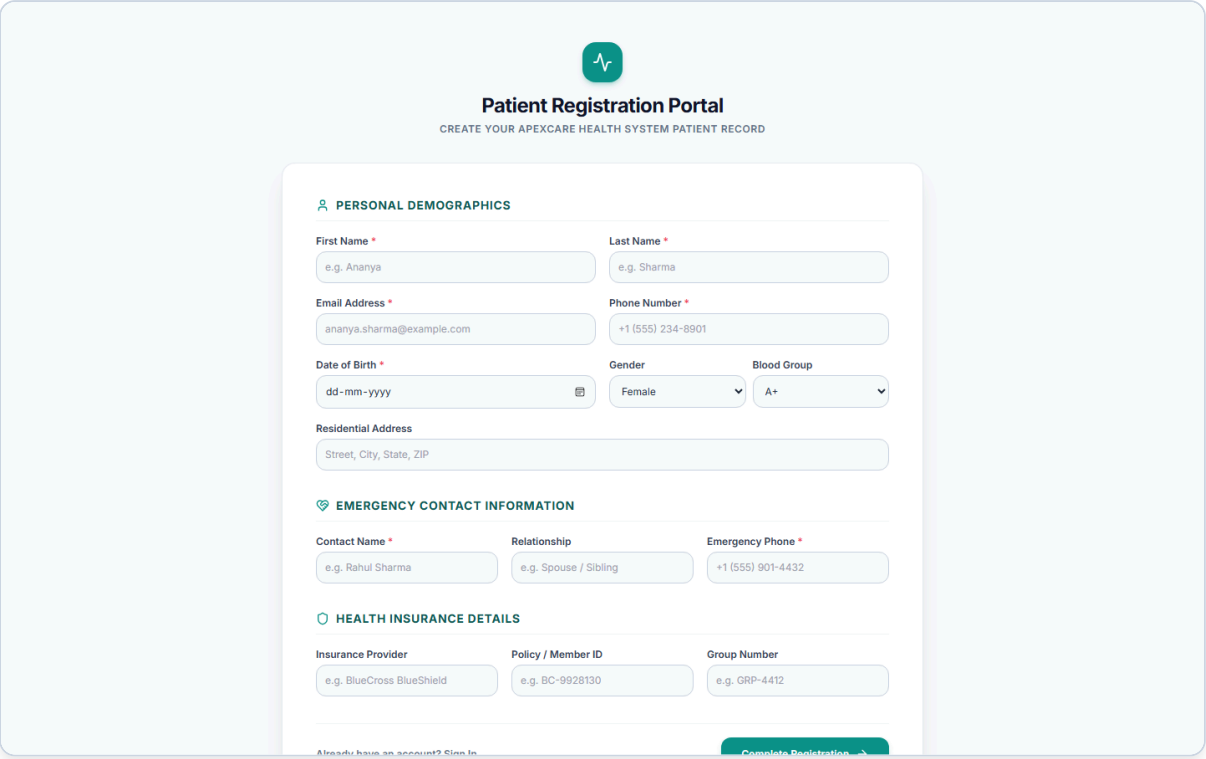
System Integration / Security Route: /login



Workflow Caption & Acceptance Evidence: The central security gateway supporting role-based access control (RBAC) across Patient, Doctor, and Administrator personas with immediate demo persona switching and secure credential authentication.

Figure 2: Patient Self-Registration & Onboarding Portal

Epic 1: Patient Registration and Demographics | Route: /register

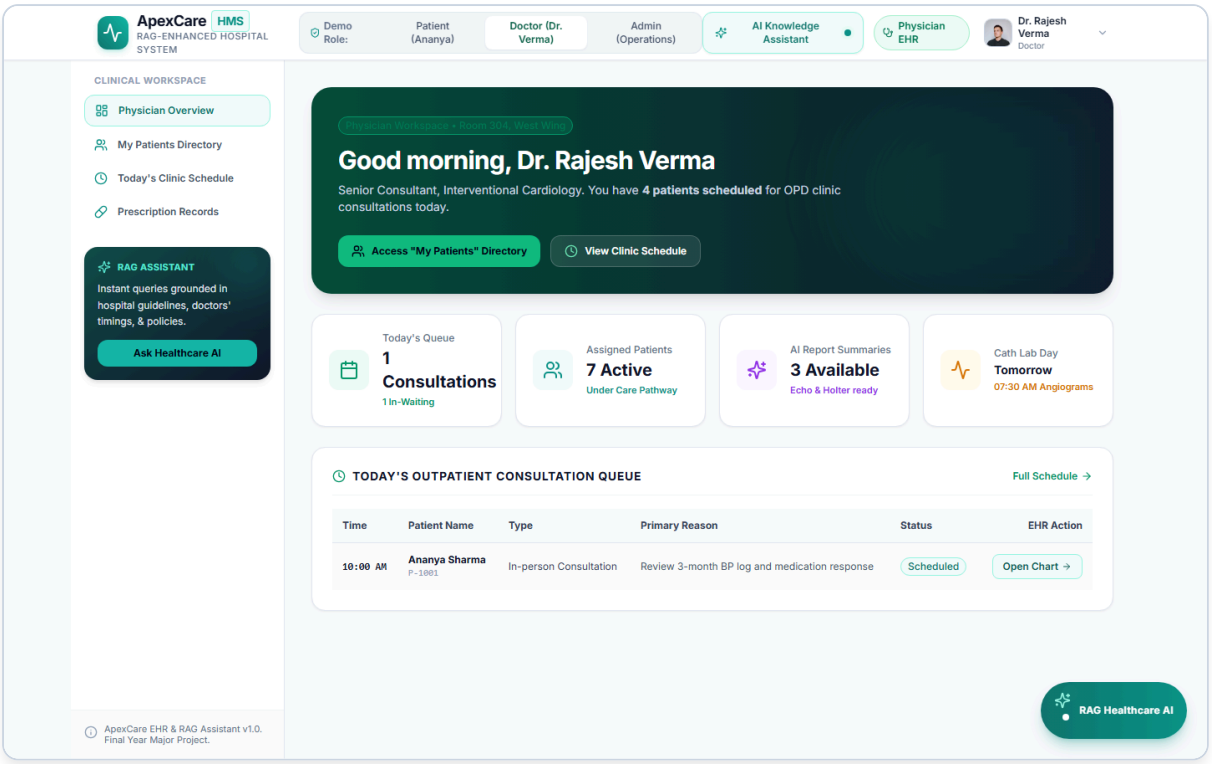


Workflow Caption & Acceptance Evidence: Captures full patient onboarding demographics including legal name, date of birth, contact information, emergency contacts, insurance policy identifiers, and initial medical history.

Figure 3: Doctor Clinical Workstation Dashboard

Epic 2: Clinical Documentation & Epic 6: System Integration

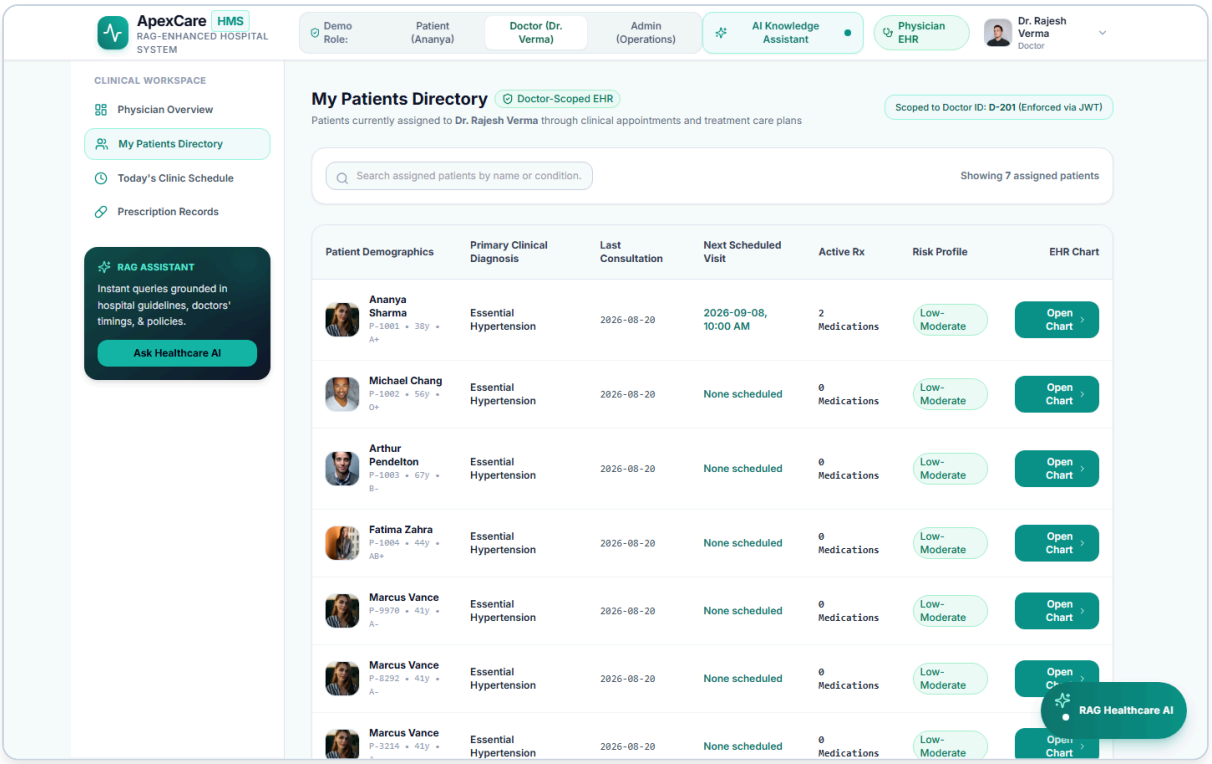
Route: /doctor/dashboard



Workflow Caption & Acceptance Evidence: Executive physician command center displaying current patient queues, critical diagnostic alerts, upcoming surgical and clinical encounters, and quick access to clinical charts.

Figure 4: Healthcare Provider Patient Directory & Search

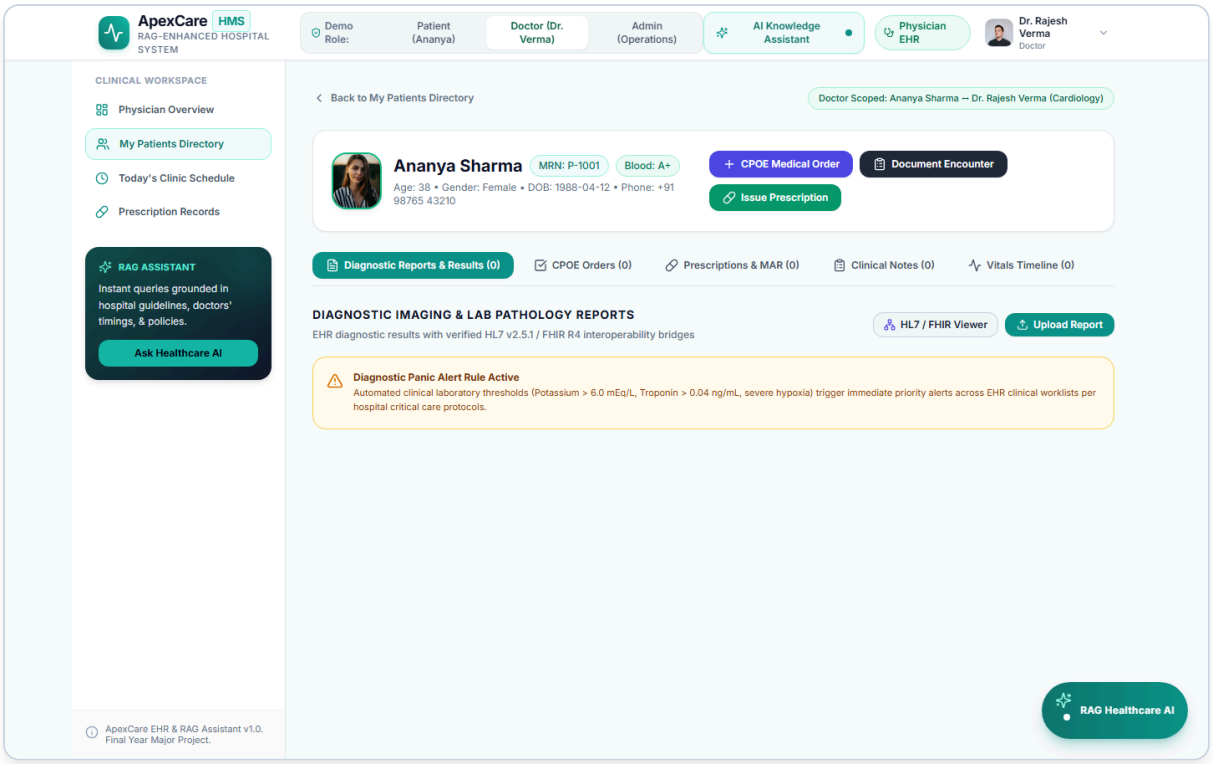
Epic 1: Patient Registration and Demographics | Route: /doctor/patients



Workflow Caption & Acceptance Evidence: Comprehensive patient directory enabling real-time filtering, patient ID lookup, active condition summaries, assigned primary physicians, and direct navigation to detailed electronic health charts.

Figure 5: Diagnostic Results Retrieval & Critical Panic Alerts

Epic 5: Lab and Imaging Integration | Route: /doctor/patients/pat-101 (Tab 1)

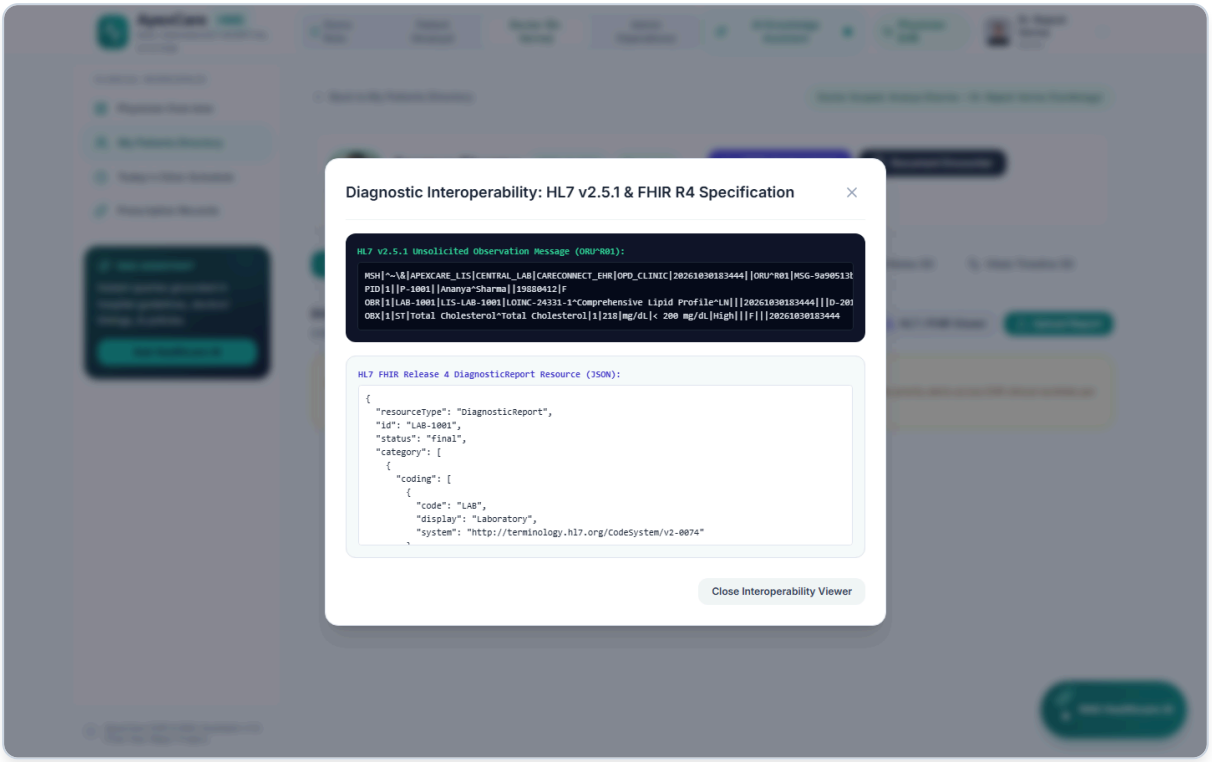


Workflow Caption & Acceptance Evidence: Unified diagnostic viewer presenting quantitative laboratory findings, reference ranges, specimen metadata, abnormal flags, and immediate red-banner panic alerts for critical electrolyte/cardiac values.

Figure 6: Diagnostic Interoperability Modal (HL7 v2.5.1 ORU^R01 & FHIR R4)

Epic 5: Lab and Imaging Integration (Interoperability)

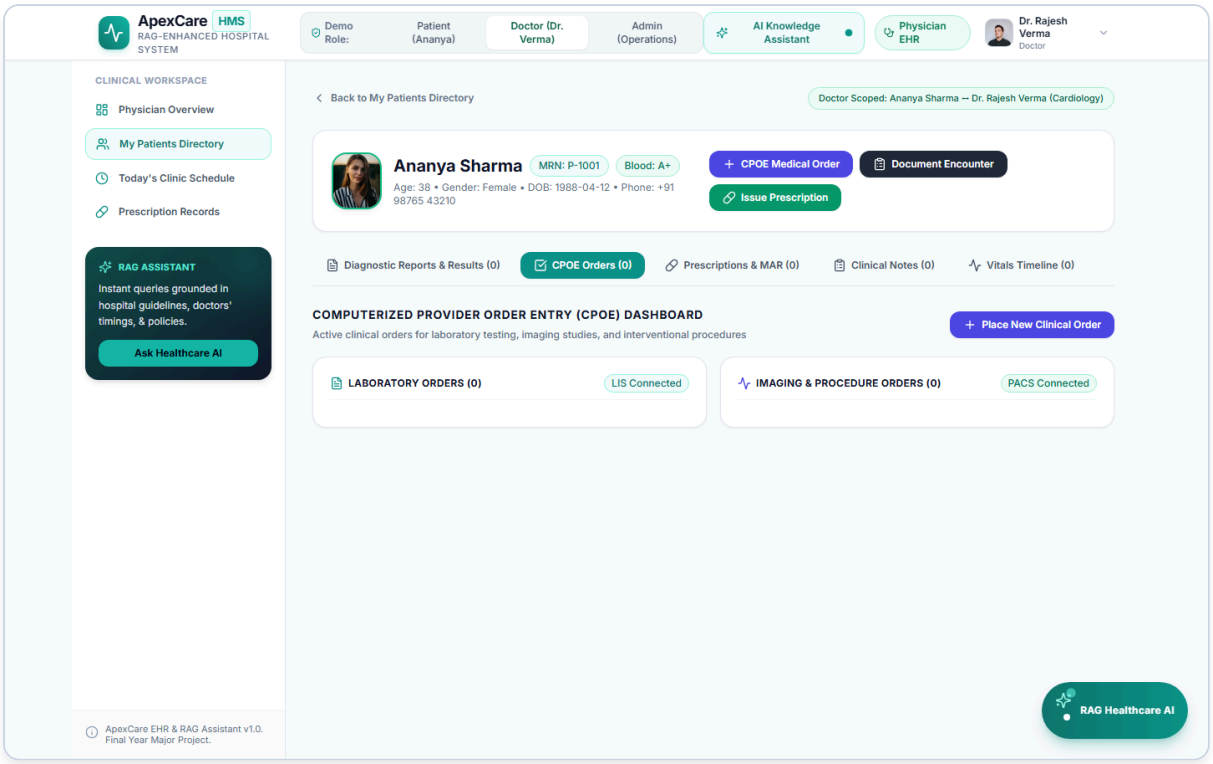
Route: /doctor/patients/pat-101 (Modal)



Workflow Caption & Acceptance Evidence: Standardized health data exchange interface generating valid HL7 v2.5.1 ORU^R01 observation segments and FHIR Release 4 DiagnosticReport/Observation JSON payloads via the CareConnect Gateway.

Figure 7: Computerized Provider Order Entry (CPOE) Dashboard

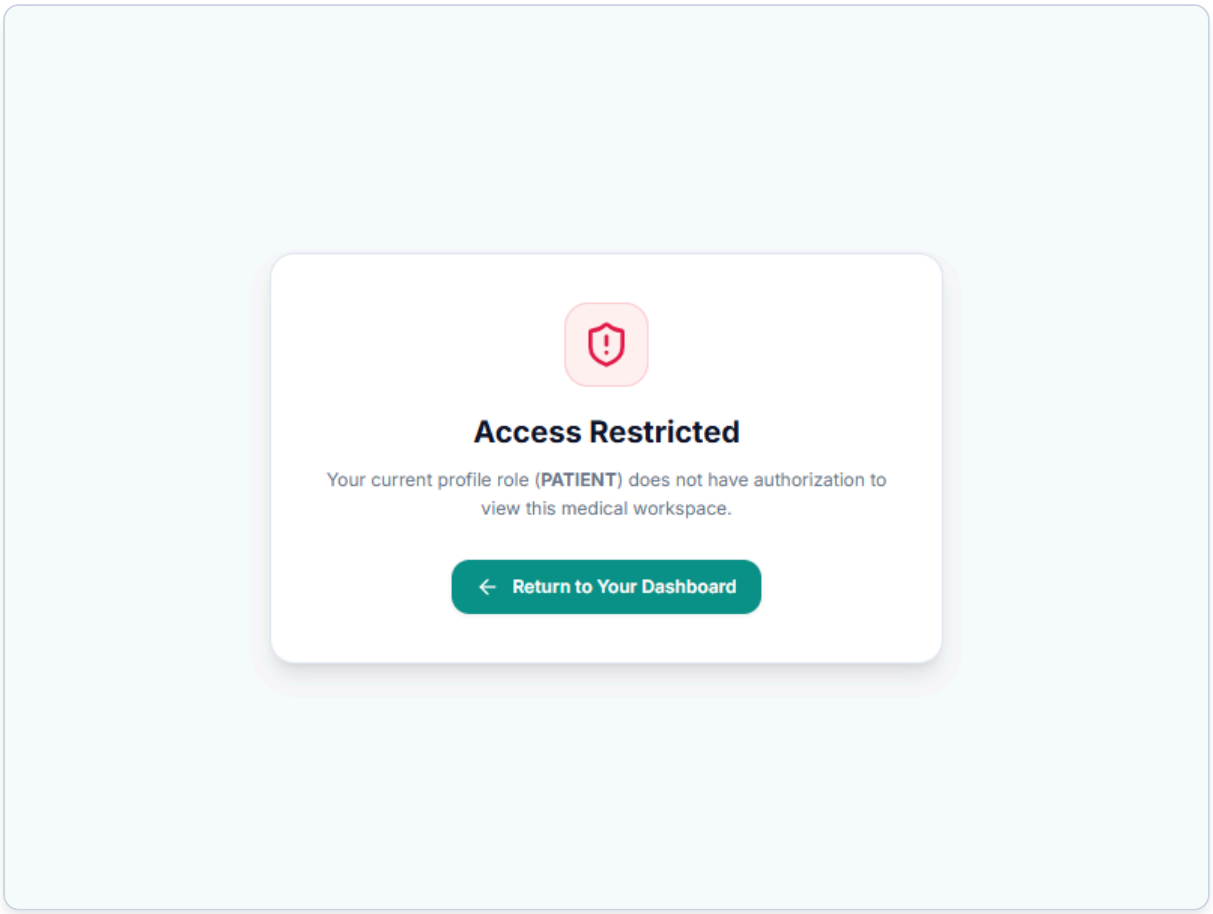
Epic 3: Computerized Provider Order Entry (CPOE) | Route: /doctor/patients/pat-101 (Tab 2)



Workflow Caption & Acceptance Evidence: Comprehensive clinical order tracking system covering laboratory tests, radiology imaging studies, and interventional procedures with order status progression (PLACED -> IN_PROGRESS -> COMPLETED).

Figure 8: CPOE Multi-Disciplinary Order Placement Interface

Epic 3: Computerized Provider Order Entry (CPOE) | Route: /doctor/patients/pat-101 (CPOE Modal)

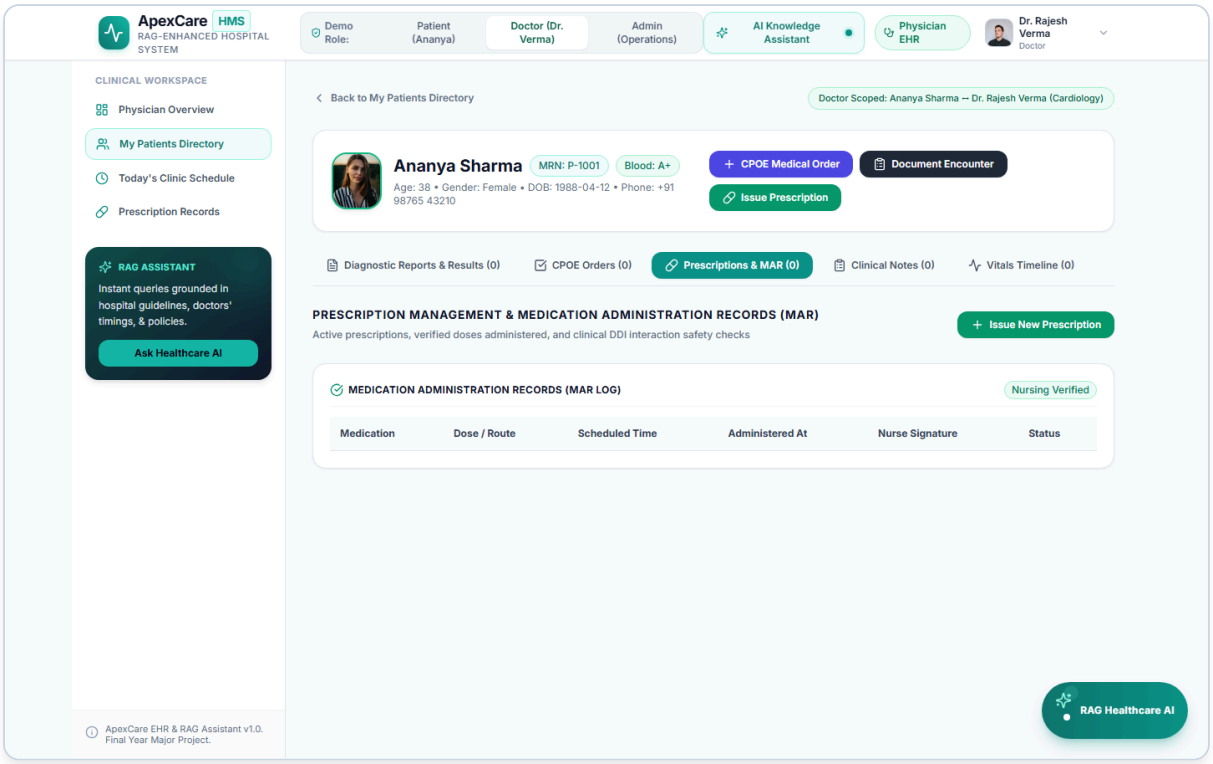


Workflow Caption & Acceptance Evidence: Interactive ordering dialog supporting standardized LOINC laboratory test selection, imaging modalities (MRI, CT, X-Ray, Ultrasound), clinical indications, and priority triage (Routine, Urgent, STAT).

Figure 9: Medication Management & MAR Administration Log

Epic 4: Medication Management

Route: /doctor/patients/pat-101 (Tab 3)

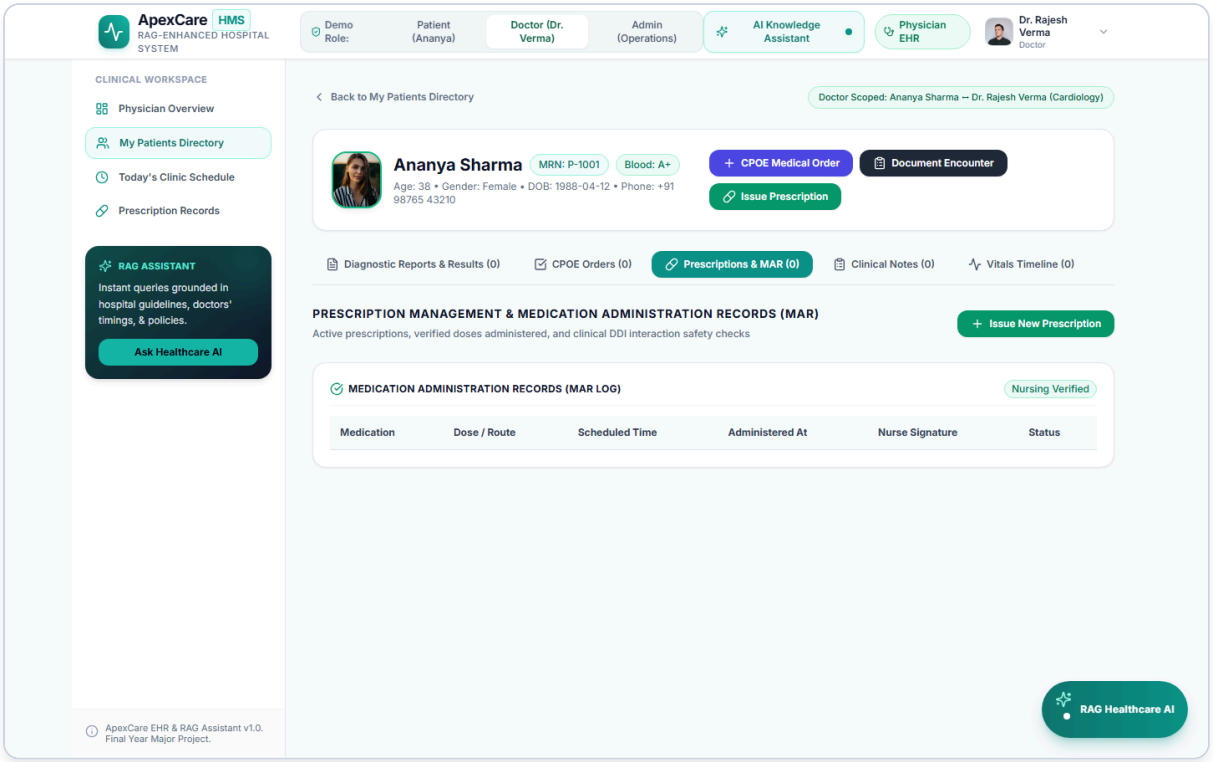


Workflow Caption & Acceptance Evidence: Electronic prescription management paired with a synchronized Medication Administration Record (MAR) tracking nurse-verified doses given, held, or refused with exact administration timestamps.

Figure 10: Drug-Drug Interaction (DDI) & Allergy Contraindication Screening

Epic 4: Medication Management (Patient Safety)

Route: /doctor/patients/pat-101 (DDI Banner)

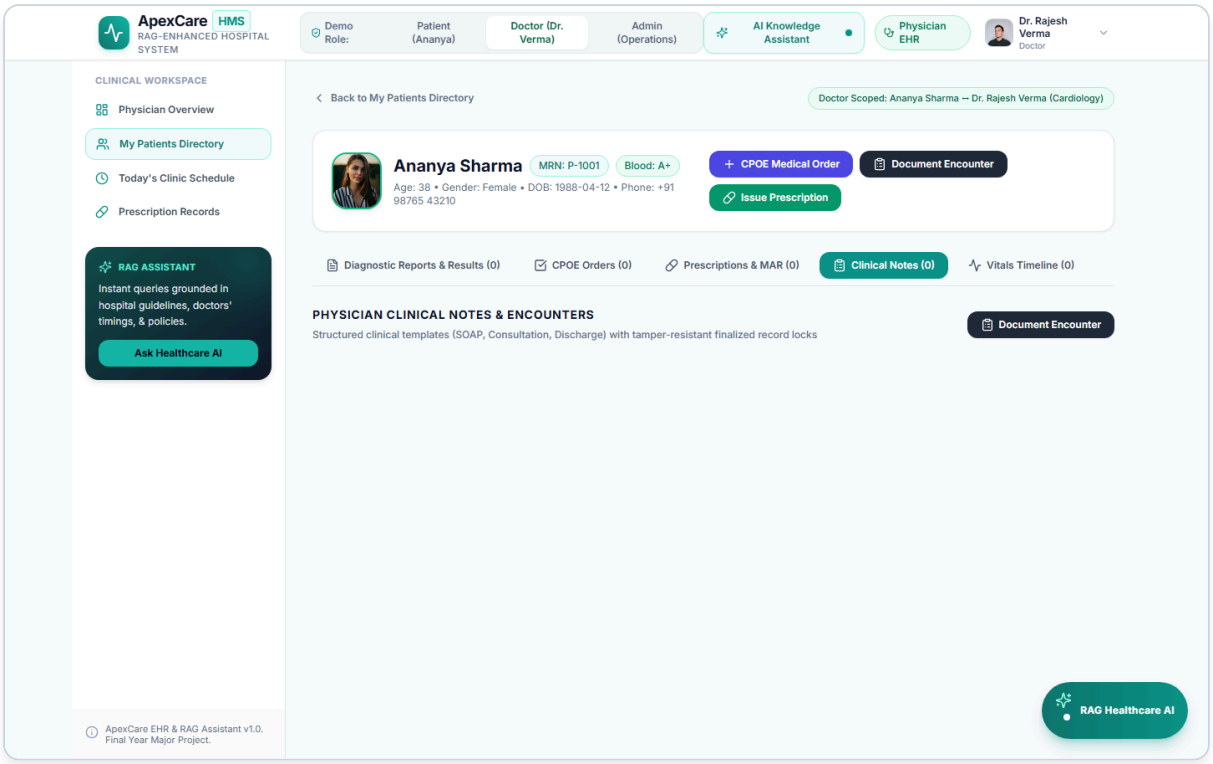


Workflow Caption & Acceptance Evidence: Integrated pharmacology safety gateway that flags severe drug-drug contraindications (e.g., Lisinopril + Spironolactone hyperkalemia risk) and allergy cross-reactions prior to prescription commit.

Figure 11: Clinical Encounter Documentation & Specialty Templates

Epic 2: Clinical Documentation

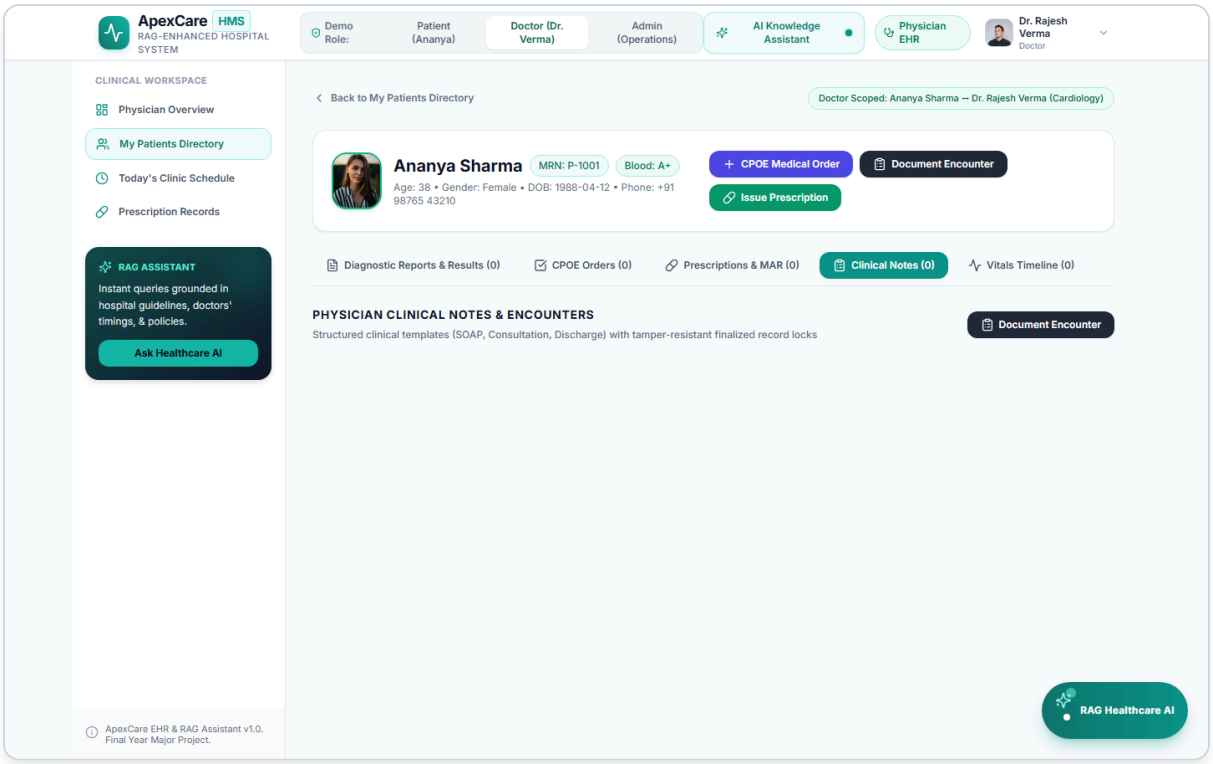
Route: /doctor/patients/pat-101 (Tab 4)



Workflow Caption & Acceptance Evidence: Provider clinical documentation workstation featuring pre-configured templates (SOAP, Cardiology Consult, Discharge Summary) with explicit separation of clinical Assessment and Treatment Plan.

Figure 12: Signed SOAP Encounter Note & Immutability Lock

Epic 2: Clinical Documentation (Tamper-Resistant EHR) | Route: /doctor/patients/pat-101 (Signed Note)

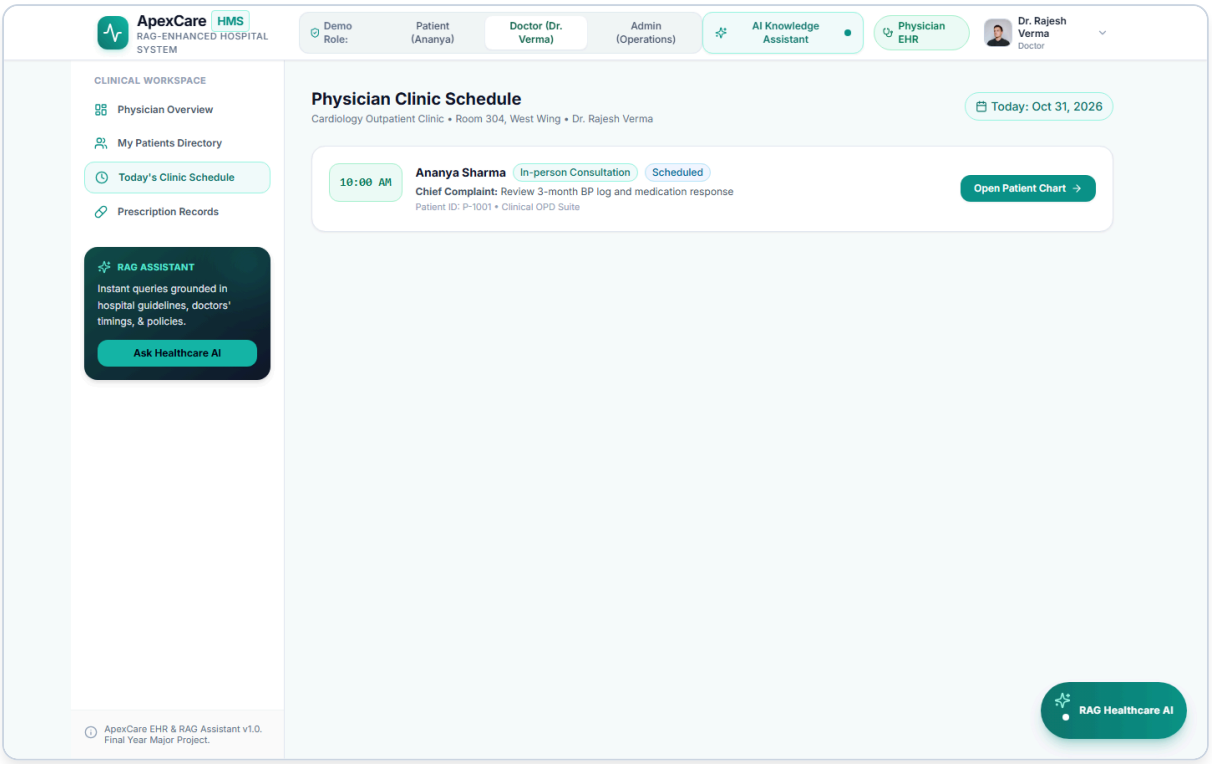


Workflow Caption & Acceptance Evidence: Demonstrates tamper-evident record locking where finalized and signed clinical notes display a locked security badge. Subsequent edit attempts are rejected by the backend with HTTP 409 Conflict.

Figure 13: Provider Schedule & Clinical Appointments Calendar

Epic 6: System Integration & Provider Scheduling

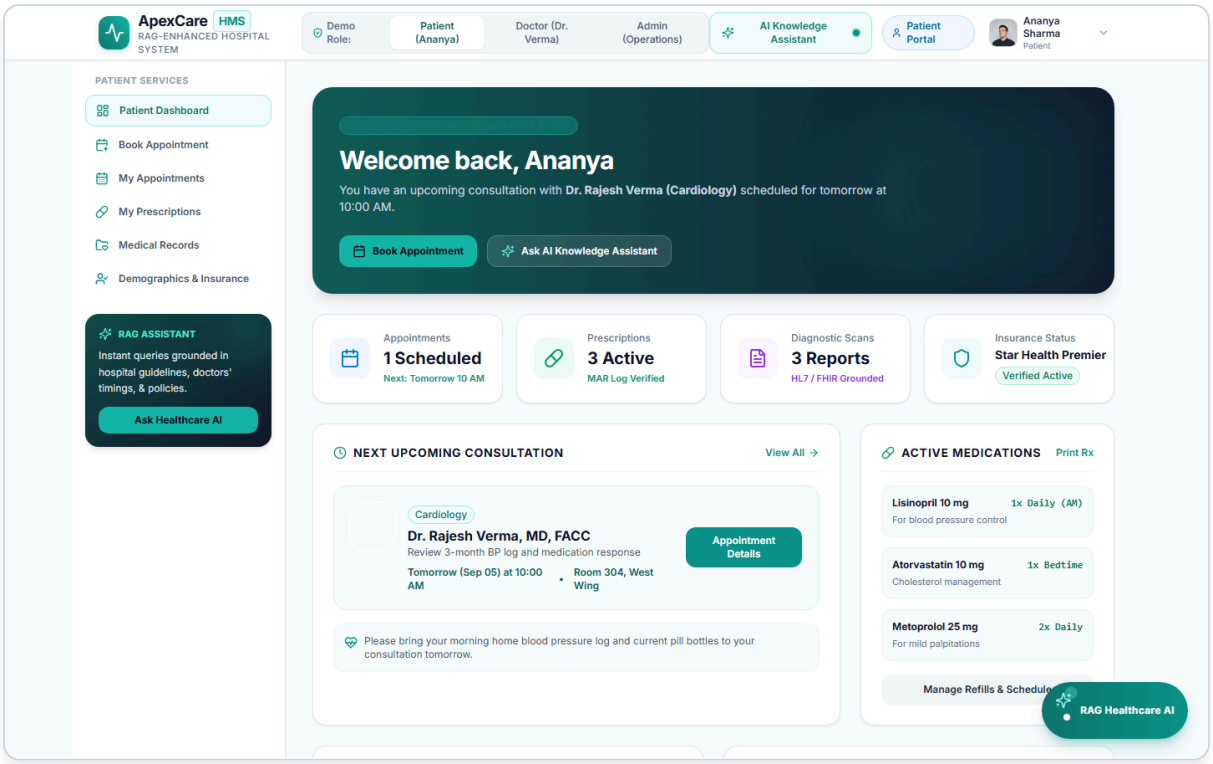
Route: /doctor/schedule



Workflow Caption & Acceptance Evidence: Provider scheduling dashboard displaying synchronized patient visits, consultation types, appointment statuses, department allocations, and encounter creation links.

Figure 14: Patient Portal Dashboard & Notifications Center

Epic 7: Patient Portal | Route: /patient/dashboard

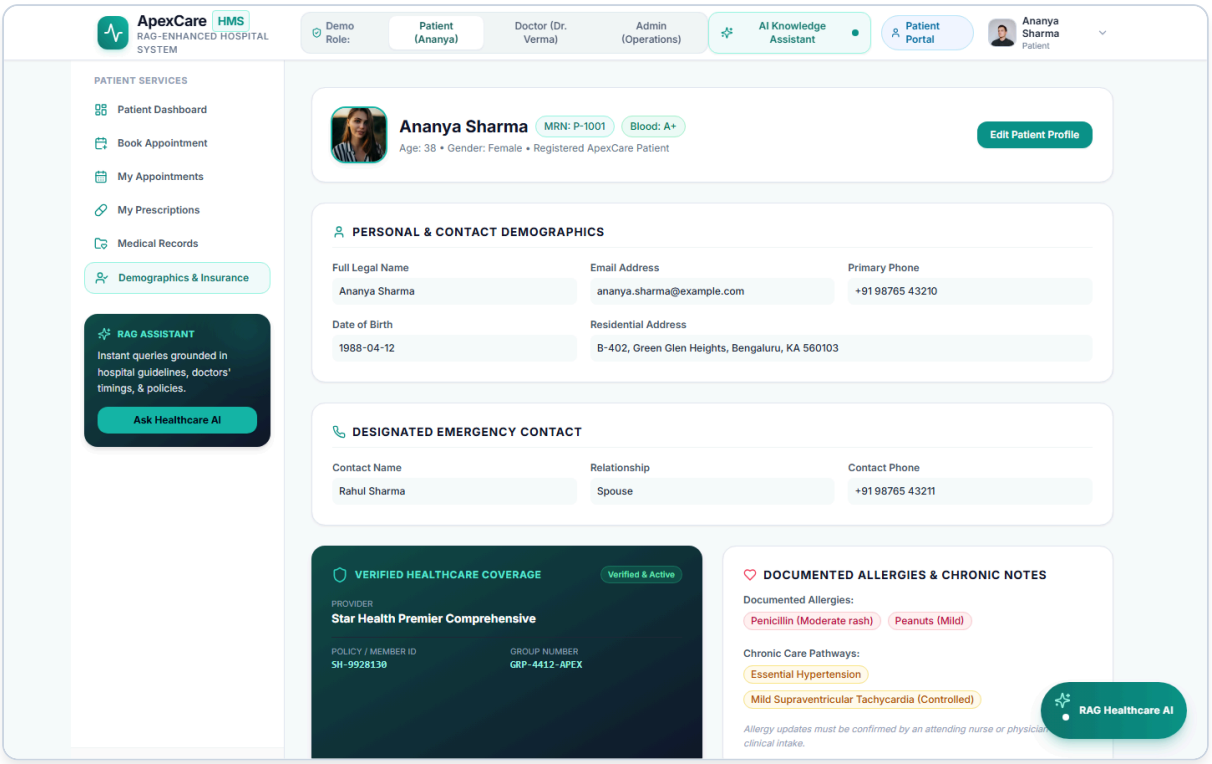


Workflow Caption & Acceptance Evidence: Patient-facing self-service hub showcasing upcoming consultations, active prescriptions, released diagnostic test results, care team contacts, and automated clinical notifications.

Figure 15: Patient Personal Health Profile & Insurance Management

Epic 1: Patient Registration & Epic 7: Patient Portal

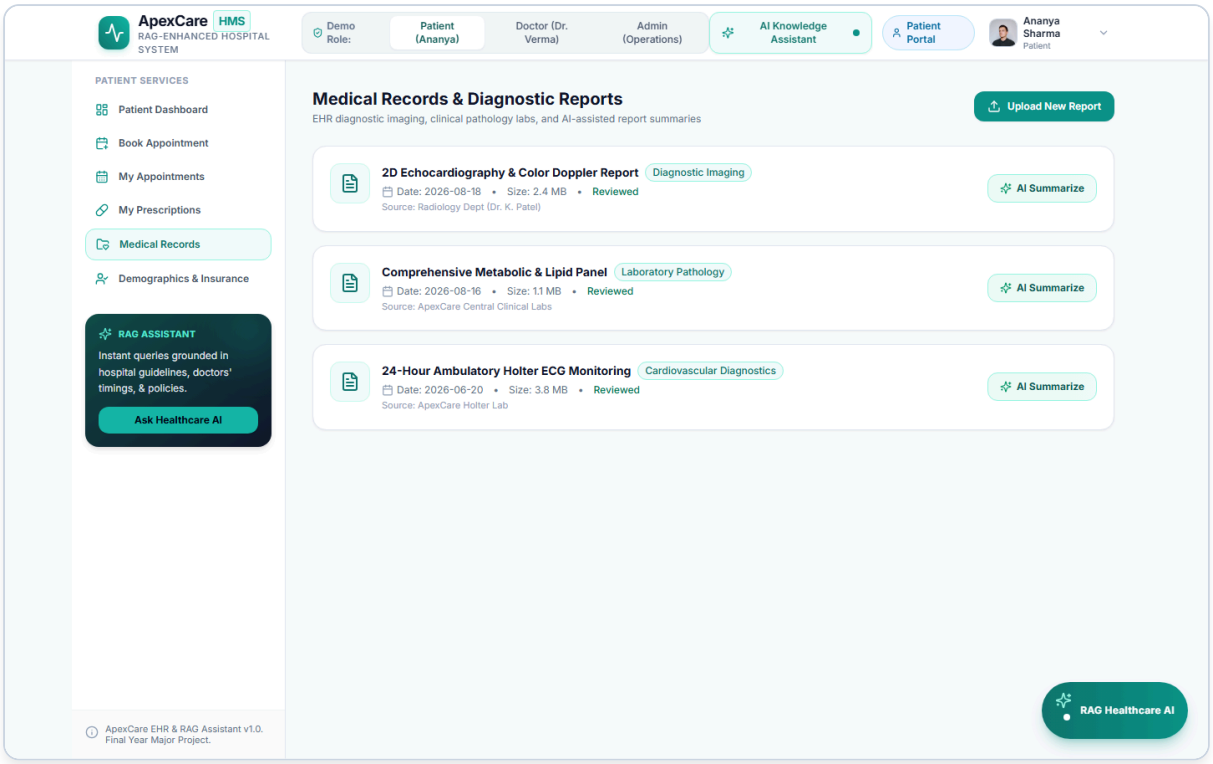
Route: /patient/profile



Workflow Caption & Acceptance Evidence: Patient demographic and coverage management interface verifying personal details, emergency contacts, primary care provider, and insurance policy coverage details.

Figure 16: Patient Self-Service Released Health Records

Epic 7: Patient Portal & Health Records Access | Route: /patient/records

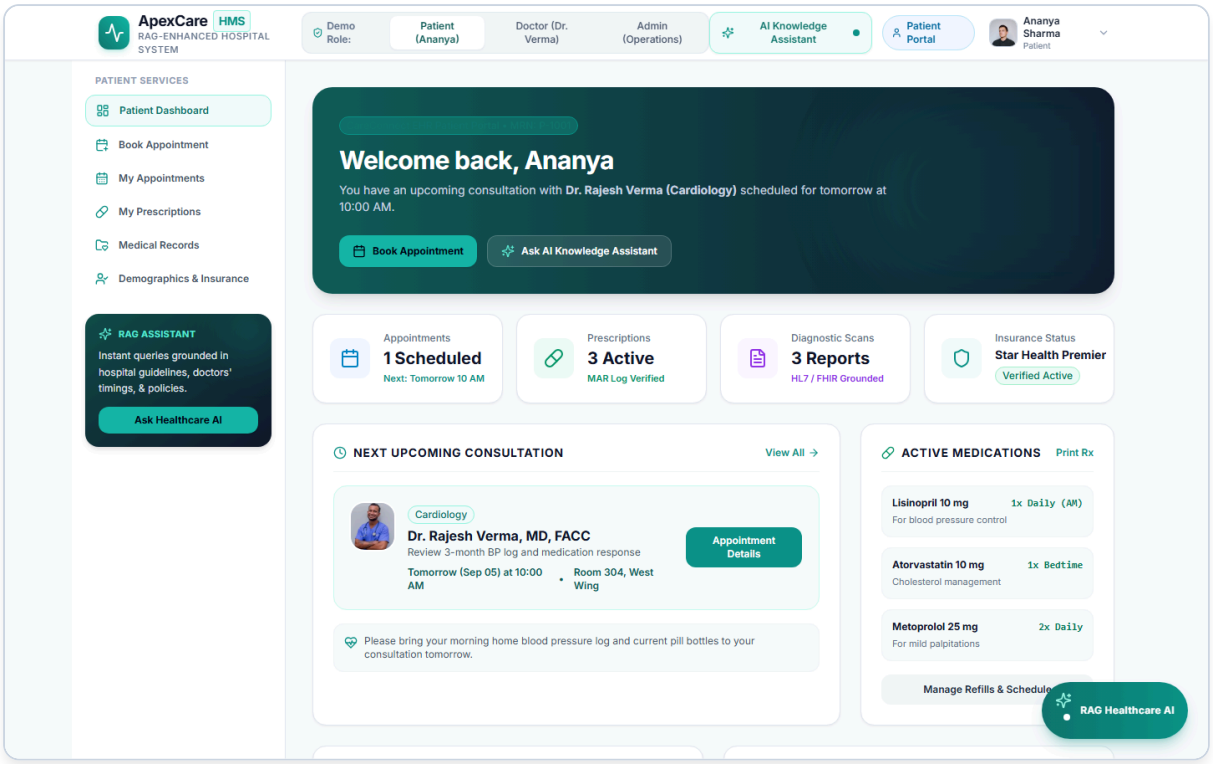


Workflow Caption & Acceptance Evidence: Secure view-only repository allowing authenticated patients to review released encounter summaries and diagnostic laboratory reports without granting write permissions to clinical notes.

Figure 17: Patient-Provider Secure Messaging & Communication

Epic 7: Patient Portal (Secure Messaging)

Route: /patient/dashboard (Messaging)

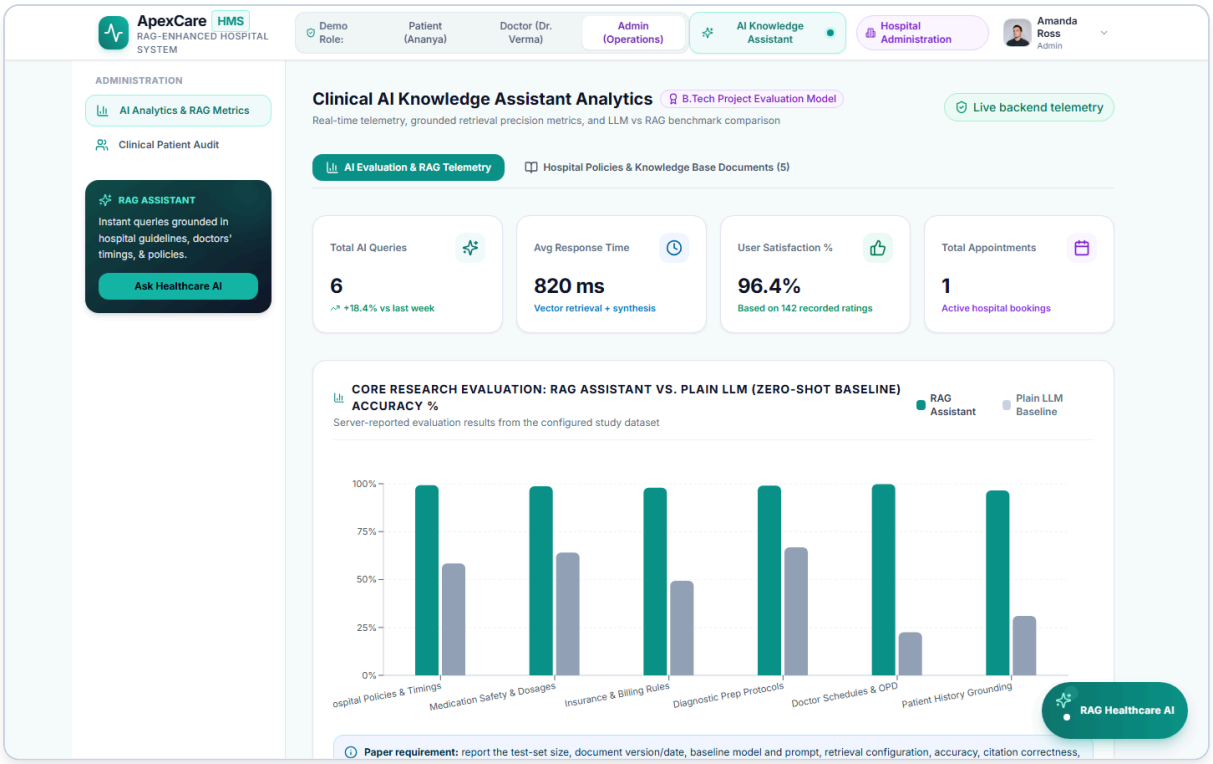


Workflow Caption & Acceptance Evidence: Confidential clinical messaging thread facilitating non-emergent patient inquiries to treating physicians with persistent conversation history and read receipts.

Figure 18: Hospital Operations Analytics & Admin Dashboard

Epic 6: System Integration & Operational Analytics

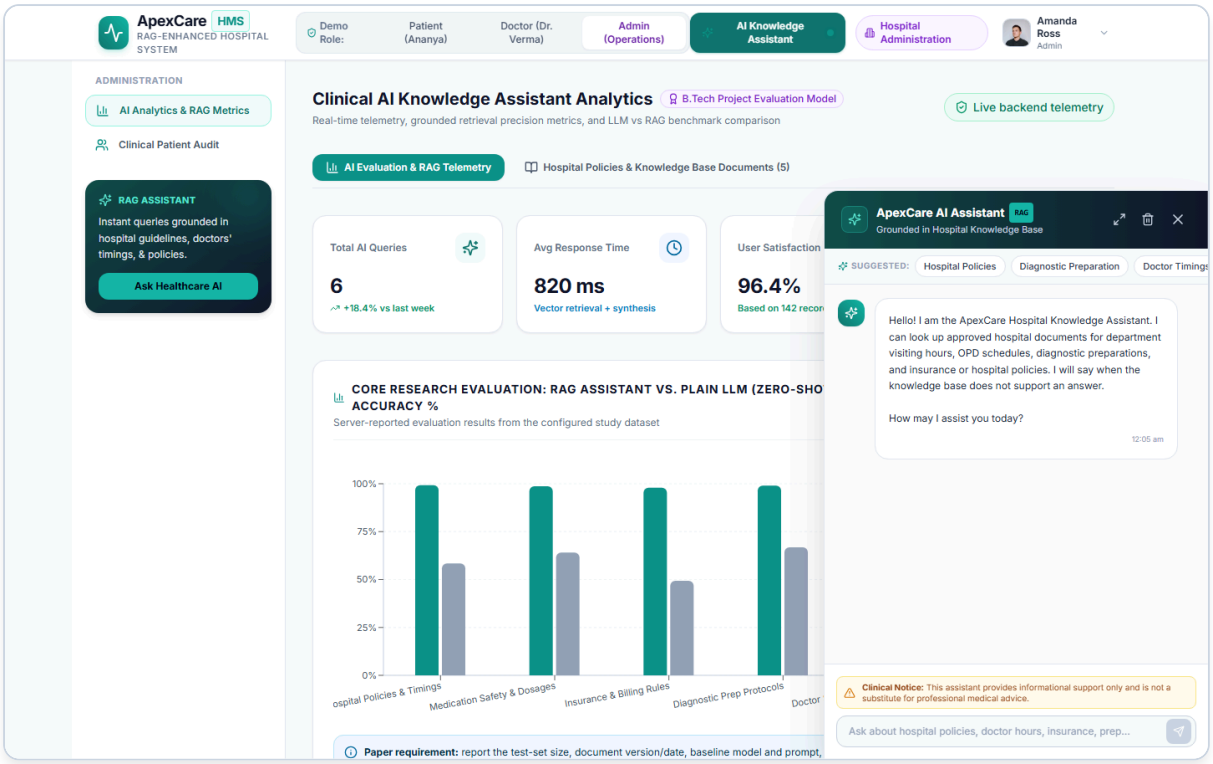
Route: /admin/analytics



Workflow Caption & Acceptance Evidence: Executive hospital analytics workstation tracking operational key performance indicators, department throughput, bed occupancy, diagnostic turnaround times, and system audit logs.

Figure 19: RAG Healthcare Knowledge Assistant with Citations

CareConnect Intelligent Assistance & Hospital Policy Search | Route: AppLayout Drawer



Workflow Caption & Acceptance Evidence: Retrieval-Augmented Generation (RAG) assistant delivering evidence-grounded answers from hospital policy documents and clinical guidelines with visible source citations and safety disclaimers.